QCOH PUSHFORWARD AND ITS DERIVED FUNCTORS COMMUTE WITH DIRECT SUMS AND FILTERED COLIMITS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Colimit cohomology. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-colimit-cohomology. Colimit commutation for f_* and R^i f_* on all modules.
- Q is exact and bicontinuous. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic (3). Q commutes with all colimits.

Source reference. Original lemma 0GQW: sources/stacks-chow/tags.txt, source lines 20876–20876. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. The functor $f_{QCoh,*}$ and the functors $R^i f_{QCoh,*}$ commute with direct sums and filtered colimits.*

Proof. The functors f_* and $R^i f_*$ commute with direct sums and filtered colimits on all modules by Lemma 0GQW. The lemma follows as $f_{QCoh,*} = Q \circ f_*$ and $R^i f_{QCoh,*} = Q \circ R^i f_*$ and Q commutes with all colimits, see Lemma 0779. □

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.